Capabilities Summary

Advanced Computational Electromagnetics & Antenna Design Consulting

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Core Engineering Services

- **Computational Electromagnetics**

  Proprietary FDTD electromagnetic simulation engine developed over 25+ years and used extensively across the defense community. Full-wave 3D analysis for antenna performance, scattering, coupling, and near-field characterization. Custom simulation tool development and RF materials characterization.

- **Antenna Design & Optimization**

  Inventor of the Fragmented Aperture Antenna (US Patents 6,323,809 & 11,228,102) — a genetically optimized antenna topology. Broadband and wideband design from UHF through millimeter wave. Phased array element design. Electrically small antennas. Custom impedance matching network design.

- **Antenna Integration Services**

  End-to-end antenna integration for IoT, wireless, and embedded devices. Antenna selection and trade studies, matching network design with component-level specs, full-wave EM simulation in device context, and test & tuning plans for prototype bring-up. LoRa, Bluetooth/Wi-Fi, cellular, GNSS, sub-GHz ISM, medical, and drones. Vendor-neutral. Remote-first. Fixed-scope engagements.

- **Platform Integration & Installed Performance**

  Installed antenna analysis on vehicles, aircraft, and ships. EMI/EMC assessment and mitigation. RCS prediction, shaping, and treatment design for low-observable platforms. Co-site interference analysis. Radome and canopy RF scanning for military specification compliance.

- **AI/ML-Accelerated Engineering (Coming August 2026)**

  ML surrogate models trained on proprietary FDTD data. Neural network-driven antenna optimization. Physics-informed ML models. AI-orchestrated design pipelines for faster iteration cycles.

- **SBIR/STTR Proposal Support**

  Technical volume writing and proposal development for SBIR/STTR grants. Innovation narratives, feasibility arguments, work plans, milestones, and solicitation alignment.

- **Patent Drafting Support**

  Technical drafting for patent materials prepared for attorney review. Invention disclosure, technical descriptions, figure preparation, and structured explanations of innovations.

Industries Served

Defense & Aerospace · Telecommunications (5G/6G) · Satellite & Space · IoT & Embedded Systems · Automotive · Intelligence Community

Products — Coming Soon

- **Fragmented Aperture Blade Antennas**
  Pre-validated stock antenna designs covering 2.4–7.5 GHz (WiFi 6E, UWB, Sub-6 5G). Available as instantly downloadable Gerber files for a fixed fee. Place directly on your product PCB or fabricate as a standalone antenna.
- **Custom Antenna Design**
  Custom fragmented aperture antenna design for specific frequency bands, substrates, or form factors. AI/FDTD optimized. Specification to validated Gerber files in 1–2 weeks.

Engagement Options

- **Initial Consultation**
  Confirm goals, scope, timeline, and deliverables (phone or virtual). No-obligation.
- **Hourly Support**
  Flexible support for ongoing tasks, technical work, and document development.
- **Fixed-Fee Projects**
  Defined scope and deliverables with agreed pricing before work begins.
- **Retainer Agreements**
  Reserved capacity for clients requiring ongoing engineering support or advisory services.

Proprietary Tools & Credentials

- **Proprietary FDTD Solver**
  Production-grade, full-wave 3D electromagnetic simulation engine. Used by multiple defense contractors and government laboratories.
- **Antenna Matching Suite**
  Specialized tools for broadband impedance matching, combining circuit synthesis with full-wave validation.
- **Patents & Publications**
  - Granted: US 6,323,809 & 11,228,102 (Fragmented Aperture) · US 11,937,270 (Flat Lens) · US 11,215,655 (Transmission Line Correction) · US 10,203,202 (Non-Contact Coating) · US 5,689,275 (Photonic Bandgap)
  - Applications: US 20030108263 & 20030108265 (Polarization Multiplexed Optical)
  - Awards: IEEE AP-S Best Paper (1993) & Runner-up (1992)
  - Book: Fragmented Aperture Antennas (in progress)
  - Scholar: scholar.google.com/citations?user=445KFCoAAAAJ

Important Notes

- Engineering consulting does not include PE-stamped plans unless explicitly contracted with a licensed PE.
- Stock antenna designs are provided as Gerber files. Integration support available separately.